



Distillation column optimization: A formal method using stage-to stage computations and distributed streams

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ABSTRACT

This work addresses the complexities of optimizing the number of stages in a distillation column, which typically lead to challenging non-linear mixed-integer optimization problems. To simplify this, we employ distributed streams, thereby eliminating discrete degrees of freedom. To avoid sophisticated initialization procedures, the optimization problem is reformulated by employing a sequence of stage-to-stage calculations, each reduced to maintaining only the MESH (mass, equilibrium, summation, heat) equations for a single stage.

Our numerical experiments show the efficiency and stability of solving the simplified optimization problem in various scenarios, including single and multiple distillation columns. For a single column scenario, we compare the accuracy of our optimization method with a full enumeration approach. Additionally, for a pressure swing flowsheet designed to separate an azeotropic mixture, we illustrate potential energy savings by optimizing a stage distribution versus using a predetermined stage distribution.

1. Introduction

Rigorous optimal design of a distillation column introduces complex optimization challenges. Employing MESH equations on each stage results in highly non-linear functions, and each stage contributes additional process variables and equations, accumulating hundreds to thousands of constraints and optimization variables for a single column. This number increases significantly when designing an entire flowsheet. Furthermore, if the optimization must determine the number of stages, discrete degrees of freedom complicate the problem further. Containing both continuous and discrete variables, these systems comprise a number of highly nonlinear features. Since distillation stages are designed to operate at vapor-liquid equilibrium (itself an optimization problem), the domain of feasible solutions may not only be limited but also difficult to determine, particularly for nonideal separations. Therefore, for systems with arbitrarily many separation stages and species to separate, as well as complex equilibrium relations, distillation optimization represents a demanding task for optimization strategies.

To simplify the optimization problem, several assumptions are typically made to the distillation model. These include constant relative volatilities or constant enthalpies, leading to constant flow rates throughout the distillation column. Initial efforts employing these re-

laxations resulted in the development of shortcut methods like those by Fenske (1932), Gilliland (1940) and Underwood (1948). These relaxations are also applicable in aggregate models where stages are grouped together. An early instance of this approach can be found in the work by Edmister (1957), with modern adaptations found in studies by Kamath et al. (2010) and Ecker et al. (2019). While these simplified models are typically easier to solve, they can lead to inaccuracies, particularly in modeling the separation of non-ideal mixtures.

In one of the first studies on optimal distillation design Sargent and Gaminibandara (1976) describe the formulation of stage optimization for distillation columns as a mixed-integer nonlinear optimization problem (MINLP). Spurred by the development of the rigorous DICOPT MINLP solver (Duran and Grossmann (1986)), Viswanathan and Grossmann (1990, 1993) formulated and solved MINLP models for optimum feed stage location and stage numbers. These studies demonstrated the capability of MINLP algorithms for multi-column and superstructure optimization, and they also enhanced the DICOPT solver with an NLP relaxation to provide better initializations, and augmented penalty functions to enable more reliable solver performance.

Several studies for multi-column systems and azeotropic separations (Bauer and Stichlmair (1998); Duennebieber and Pantelides (1999)) also consider the formulation of stage optimization of distillation columns as

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mixed-integer nonlinear programs (MINLP). In addition, a disjunctive programming approach was introduced for these MINLP formulations (Yeomans and Grossmann (2000)). This extension was shown to be more versatile in Barttfeld et al. (2003) and applicable to complex problems, such as membrane-assisted distillation processes (Skiborowski et al. (2014)). However, it is important to note that in general, MINLPs may still be challenging to solve.

One strategy to avoid solving the discrete and continuous parts of the optimization problem simultaneously is to separate them. First, the discrete degrees of freedom are fixed, allowing the optimization to focus solely on the continuous degrees of freedom. Subsequently, the discrete degrees are optimized in an outer loop. Provided the continuous part of the problem can be solved robustly, various options are available for handling the discrete component. For instance, an evolutionary algorithm for optimizing the outer loop was used for a reactive distillation column in Urselmann and Engell (2015). Alternatively, in Palma-Flores et al. (2023), a discrete steepest descent algorithm was employed, considering both the design and nonlinear model predictive control under uncertainty simultaneously.

In contrast, MINLP-based distillation optimization problems were simplified in Lang and Biegler (2002) by replacing the integer variables with continuous ones, using differentiable distribution functions. This determines the placement of feed and product stages, while the optimal number of stages is obtained from the distribution of the reflux stream in the column. In the same vein, Dowling and Biegler (2014) introduce bypass streams for each stage to determine the fraction of the stream bypassing a stage; this approach was further incorporated into a framework with an initialization scheme (Dowling and Biegler (2015)). To coerce the continuous values toward either 0 or 1, penalty functions were introduced in Liu et al. (2022) and a different relaxation in Jia et al. (2022). Additionally, Jia et al. (2023) describe a procedure for adaptively adding stages to ensure that stages, which are not bypassed, remain consecutive. Moreover, to ease initialization of NLPs, Pattison and Baldea (2014) proposed a pseudo-transient (i.e., homotopy-continuation) approach for robust simulation and optimization of distillation models.

In the present paper, we focus on rigorously solving the design problem for both single and multiple distillation columns. A common requirement for the approaches previously described is the necessity for careful initialization. For instance, a base-case simulation problem often serves as an initialization to the optimization problem. While this can be a reliable and efficient strategy, our approach presents an alternative that is independent of a particular (often problem dependent) initialization strategy. In Seidel et al. (2020), stage-to-stage calculations were utilized to solve the majority of the MESH equations, allowing the optimization problems to be solved without special initialization procedures. This approach was further applied to dividing wall columns and multiple dividing wall columns in Seidel et al. (2022). To accommodate variable-sized columns in our model, we will employ distribution functions introduced in Lang and Biegler (2002). We extend the stage-to-stage calculations to this new model and demonstrate that they can be performed consistently. The stage-to-stage calculation will be the key to a robust optimization without tedious initialization.

We present results of two numerical studies to validate the robustness of our approach. Initially, we apply our methodology to the design of a single distillation column. With a predetermined separation task, we aim to minimize the number of stages required for the separation. Remarkably, the problem could be solved robustly across a broad range of parameters, even without an initialization procedure. A comparison of our results with those obtained from a time-intensive full enumeration confirms good agreement. Subsequently, we explore flowsheets comprising multiple distillation columns, focusing specifically on the pressure swing distillation of an azeotropic mixture. In this scenario, we examine the energy minimization between an optimal distribution of stages versus a fixed stage distribution.

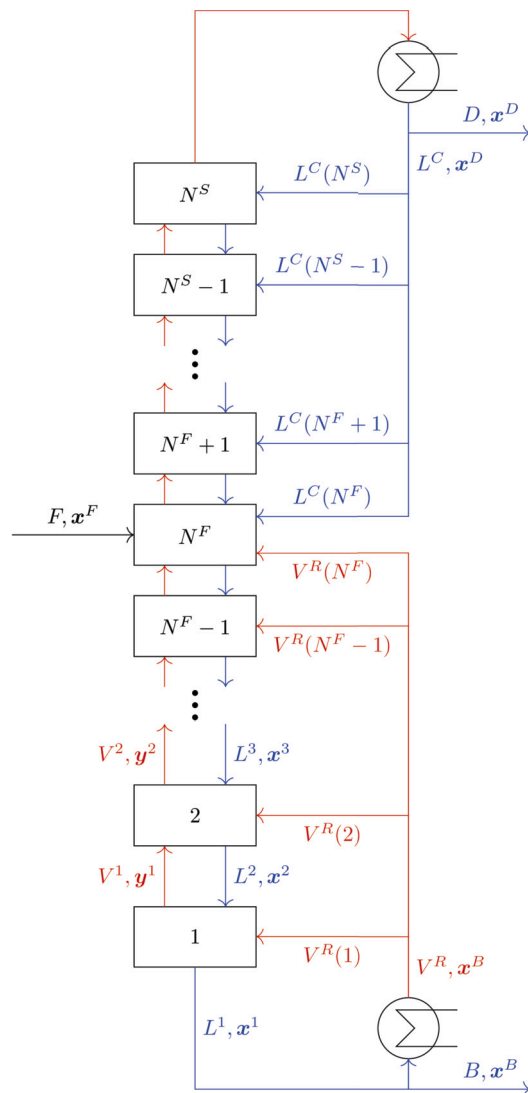


Fig. 1. Equilibrium stage model with variable number of stages in rectification and stripping section.

The paper is structured as follows. The subsequent section details the model incorporating distributed streams. In Section 3, we introduce the stage-to-stage calculations and discuss their guaranteed evaluation. The methodology is then applied to two scenarios in the following sections: initially to a single column design problem, and subsequently to flowsheets containing multiple columns.

2. Model

We model a distillation column in steady state using the equilibrium stage model with at most N^S stages. The schematic is shown in Fig. 1.

The column enters a feed stream on stage N^F , which consists of N^C different components. The flow rate of the feed stream is denoted as F , and the molar composition of each component c is represented by x_c^F . We can collect all component compositions in a vector $\mathbf{x}^F$. The temperature of the feed stream is denoted by T^F .

For each stage n , there is a liquid stream with flow rate L^n and molar composition vector $\mathbf{x}^n$ flowing from it to the stage below, and a vapor stream with flow rate V^n and molar composition vector $\mathbf{y}^n$ flowing from it to the stage above. Both streams have the temperature T^n . We assume a constant pressure p throughout all stages in the column. However, the following can easily be extended to consider variable pressure using for example a pressure drop correlation.

A portion of the liquid stream at the bottom is withdrawn as the bottom product stream with flow rate B , liquid composition $\mathbf{x}^B$ and temperature T^B . The remaining liquid is vaporized in the reboiler with a reboiler duty of Q^R , producing a vapor stream with flow rate V^R and temperature T^R . In contrast to the standard model for a column with a fixed number of stages, this stream doesn't enter a single stage, but is distributed over stages 1 to N^F . This way we can account for a variable number of stages in the stripping section. The flow to each stage n is denoted as $V^R(n)$, satisfying the equation:

$$\sum_{n=1}^{N^F} V^R(n) = V^R.$$

At the top of the column, the vapor stream from stage N^S is completely condensed, using the negative condenser duty Q^C and then split into the top product stream with flow rate D , composition $\mathbf{x}^D$ and temperature T^D . The remaining stream has a flow rate L^C and temperature $T^C = T^D$. To account for a variable number of stages in the rectification section, the target of this stream is distributed over stages N^F to N^S . The flow to each stage n is denoted as $L^C(n)$, satisfying the equation:

$$\sum_{n=N^F}^{N^S} L^C(n) = L^C.$$

We assume that the streams leaving the reboiler and the condenser are at dew point and bubble point, respectively. This means that the temperature of the streams can be calculated using their composition vectors. Throughout the column, we assume that the MESH equations hold in each stage. Exemplarily, the equations are given for a stage $2 \leq n \leq N^F - 1$ by:

$$\begin{aligned} V^n \mathbf{y}^n + L^n \mathbf{x}^n &= V^{n-1} \mathbf{y}^{n-1} + L^{n+1} \mathbf{x}^{n+1} + V^R(n) \mathbf{x}^B, \\ V^n h^V(\mathbf{y}^n, T^n) + L^n h^L(\mathbf{x}^n, T^n) &= V^{n-1} h^V(\mathbf{y}^{n-1}, T^{n-1}) \\ &+ L^{n+1} h^L(\mathbf{x}^{n+1}, T^{n+1}) \\ &+ V^R(n) h^V(\mathbf{x}^B, T^R), \\ \mathbf{y}^n &= \mathbf{K}(\mathbf{x}^n, \mathbf{y}^n, T^n, p) \mathbf{x}^n, \\ \sum_{c=1}^{N^C} x_c^n &= \sum_{c=1}^{N^C} y_c^n = 1, \end{aligned}$$

where $h^V(\mathbf{y}, T)$ denotes the vapor enthalpy with composition $\mathbf{y}$ and temperature T , and $h^L(\mathbf{x}, T)$ denotes the liquid enthalpy with composition $\mathbf{x}$ and temperature T . The values $\mathbf{K}(\mathbf{x}, \mathbf{y}, T, p)$ denote the equilibrium ratios. The full set of equations is given in Appendix A.

To simplify the exposition in the following we introduce the notation $\hat{h}^V(\mathbf{y}) := h^V(\mathbf{y}, T)$ as the vapor enthalpy of $\mathbf{y}$ with the dew point temperature T and $\hat{h}^L(\mathbf{x}) := h^L(\mathbf{x}, T)$ as the liquid enthalpy of $\mathbf{x}$ at bubble point temperature T . The enthalpy equation then reads:

$$V^n \hat{h}^V(\mathbf{y}^n) + L^n \hat{h}^L(\mathbf{x}^n) = V^{n-1} \hat{h}^V(\mathbf{y}^{n-1}) + L^{n+1} \hat{h}^L(\mathbf{x}^{n+1}) + V^R(n) \hat{h}^V(\mathbf{x}^B).$$

To model a specific number of stages in the stripping section, we can assume that $V^R(n)$ is 0 for all stages except for one stage $n' \in [1, N^F]$. If $n' > 1$, then stage 1 is thermodynamically equivalent to a flash, with a liquid input stream from the stage above and no entering vapor stream. Consequently, the output of this stage will be purely liquid, resulting in $V^1 = 0$, $L^1 = L^2$, and $\mathbf{x}^1 = \mathbf{x}^2$. This argument can be repeated for all stages n with $n < n'$. As a result, the stripping section is equivalent to the stripping section of a column that doesn't have the first $n' - 1$ stages. Analogously, we can model a specific number of stages in the rectification section by setting all $L^C(n)$ values to zero except for one stage.

However, allowing only one non-zero $V^R(n)$ introduces a discrete degree of freedom, and no derivative with respect to n' exists. To avoid

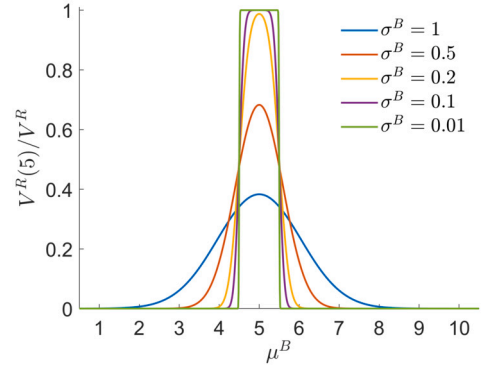


Fig. 2. Relative portion $\frac{V^R(5)}{V^R}$ of vapor entering stage 5 as a function of μ^B for different values of σ^B with feed stage $N^F = 10$.

solving a mixed-integer nonlinear optimization problem, we can introduce a Gaussian distribution over the position, as it was also done in Lang and Biegler (2002). We define for $1 \leq n \leq N^F$:

$$V^R(n) = \frac{\Phi_{\mu^B, \sigma^B}(n + 0.5) - \Phi_{\mu^B, \sigma^B}(n - 0.5)}{\Phi_{\mu^B, \sigma^B}(N^F + 0.5) - \Phi_{\mu^B, \sigma^B}(0.5)} V^R, \quad (1)$$

where $\Phi_{\mu^B, \sigma^B}(\cdot)$ denotes the cumulative distribution function of the Gaussian distribution with mean $\mu^B \in [1, N^F]$ and standard deviation $\sigma^B > 0$, i.e. for $t \in \mathbb{R}$:

$$\Phi_{\mu^B, \sigma^B}(t) = \int_{-\infty}^t \frac{1}{\sigma^B \sqrt{2\pi}} \exp\left(-\frac{1}{2} \left(\frac{x - \mu^B}{\sigma^B}\right)^2\right) dx.$$

This means that we use the weight of the Gaussian distribution in the interval $[n - 0.5, n + 0.5]$ and assign this portion of V^R to the stage n . As we can only distribute the stream over the stages 1 to N^F , we need to make sure that the portions sum up to V^R . This is achieved by normalizing the weights with $\Phi_{\mu^B, \sigma^B}(N^F + 0.5) - \Phi_{\mu^B, \sigma^B}(0.5)$.

The idea is as follows: If we fix the mean μ^B and let σ^B approach zero, we converge towards the case where $V^R(n)$ is identical to V^R for $\mu^B \in [n - 0.5, n + 0.5]$ and 0 otherwise. On the other hand, for every $\sigma^B > 0$ and stage n , the vapor flow rate $V^R(n)$ is differentiable with respect to μ^B . This approach can thus be seen as smoothing the case of a discrete number of stages. By adjusting σ^B , we can control how close to the discrete case and “how smooth” the function $V^R(n)$ is with respect to μ^B . The function for different values of σ^B is shown in Fig. 2. For small values of σ^B the description is closer to the discrete case, but also introduces steep gradients. In the numerical examples presented in Sections 4 and 5, we will choose $\sigma^B = 1$ and show that this already leads to a good agreement with the discrete case.

Analogous to the case of the stripping section, for a stage $N^F \leq n \leq N^S$, we let the liquid flow rate from the condenser be

$$L^C(n) = \frac{\Phi_{\mu^D, \sigma^D}(n + 0.5) - \Phi_{\mu^D, \sigma^D}(n - 0.5)}{\Phi_{\mu^D, \sigma^D}(N^S + 0.5) - \Phi_{\mu^D, \sigma^D}(N^F - 0.5)} L^C,$$

where $\mu^D \in [N^F, N^S]$ and $\sigma^D > 0$.

For small σ^B and σ^D , the vapor streams V^n for n below μ^B and the liquid streams L^n for stages n above μ^D will be close to 0. This is why we define the number of stages that effectively describe the column as:

$$\eta = \mu^D - \mu^B + 1.$$

To simplify the notation, we introduce the following variables:

$$\begin{aligned} \bar{V}^R(n) &:= V^R - \sum_{k=1}^n V^R(k), \quad \text{for } n \leq N^F, \\ \bar{L}^C(n) &:= L^C - \sum_{k=n}^{N^S} L^C(k), \quad \text{for } n \geq N^F. \end{aligned}$$

These represent the amount of vapor from the reboiler entering the column above stage n and the amount of liquid from the condenser entering the column below stage n .

3. Model-reduction using stage-to-stage

Before describing the detailed reduction approach, we introduce and describe the general idea, following the principle introduced in Seidel et al. (2020).

We collect all process variables described above (i.e. flow rates, composition vectors, temperatures and heat duties) in $\mathbf{w}$ and the MESH equations by $h_i, i \in I_{\text{MESH}}$. The process variables describe a distillation column in steady state if:

$$h_i(\mathbf{w}) = 0 \quad \text{for all } i \in I_{\text{MESH}}.$$

These equations will form the basis for the next sections, where we discuss the design of a single distillation column and then turn to optimizing a flowsheet based on multiple distillation columns. Before solving these models, we aim to simplify the equations. The idea is to choose a small set of decision variables $\mathbf{z}$ and to solve a sequence of MESH equations, starting from the bottom and the top and moving toward N^F for all process variables. This allows us to map the degrees of freedom to the process variables:

$$\mathbf{z} \mapsto \mathbf{w}. \quad (2)$$

As a result, we ensure that the majority of the MESH equations are already satisfied and can thus be omitted. This means that we only need to solve a reduced system:

$$h_i(\mathbf{w}(\mathbf{z})) = 0 \quad \text{for all } i \in I_{\text{red}},$$

where $I_{\text{red}} \subseteq I_{\text{MESH}}$ correspond to the MESH equations at the feed stage. For this approach to be valid, we need to ensure that the map $\mathbf{z} \mapsto \mathbf{w}$ is well-defined for all possible values of $\mathbf{z}$. Additionally, we need a robust approach to evaluate this map and, as a result, pre-solve a majority of the MESH equations.

On the one hand, the goal is to choose significantly fewer optimization variables $\mathbf{z}$ than the number of process variables $\mathbf{w}$. This reduction of most of the MESH equations leads to a smaller problem that is easier to solve. On the other hand, we will choose more optimization variables than the degrees of freedom needed to fully specify a distillation column. This means that we do not solve all MESH equations and, in this sense, do not perform a full simulation of a distillation column. However, we can solve the majority of them. The equations that we solve in the reduction are chosen in such a way that we can solve them reliably and fast. In fact, we will provide guarantees and a safeguarded method for the reduction in this section. This decomposition will enable us to simulate and optimize simultaneously without difficult initialization procedures.

The proposed solution approach can be seen as an iterative procedure with an outer part (optimization) and inner part (reduction). The outer part suggests values for the variables $\mathbf{z}$, while the inner part evaluates the map given in (2). The outer part can then check the values of the reduced system. The outer part will be done by formulating the remaining equations as constraints for the optimization problem. The optimization routine will then suggest values for $\mathbf{z}$, which are used in the inner part. This approach allows us not only to solve the remaining equations but also to simultaneously minimize an objective and respect additional constraints. We will introduce the specific optimization problems in Section 4 and Section 5. In the remaining part of this section, we will focus on the inner procedure.

After this general description, we now turn to our concrete model introduced in Section 2. As optimization variables $\mathbf{z}$, we consider the following:

- the bottom flow rate $\dot{m}_c^B := Bx_c^B$ of each component $1 \leq c \leq N^C$.
- the distillate flow rate $\dot{m}_c^D := Dx_c^D$ of each component $1 \leq c \leq N^C$.

- the mean positions of the streams returning to the column μ^B, μ^D .
- the flow rate of the vapor leaving the reboiler and the liquid leaving the condenser V^R, L^C .

These variables are fixed prior to the reduction. As described above, the optimization routine will adjust them in the outer loop. Additionally, we will assume that the pressure p and the standard deviations σ^B and σ^D are fixed. Unlike the optimization variables, we consider p, σ^B and σ^D fixed for the optimization in the outer loop as well. Using these values, we can calculate all process variables $\mathbf{w}$. In this procedure, we can ensure that all MESH equations in the stripping section (i.e., stages strictly below the feed stage N^F) and rectification section (i.e., stages strictly above the feed stage) will be satisfied. The only remaining equations that we need to solve in the outer loop are the equations for the feed stage N^F . Since we include these remaining equations in the optimization problem, the number of optimization variables is larger than the degrees of freedom for a distillation column.

We start with the distillate and bottom streams. We can compute the flow rates and the compositions by:

$$B = \sum_{c=1}^{N^C} \dot{m}_c^B, \quad (3)$$

$$D = \sum_{c=1}^{N^C} \dot{m}_c^D, \quad (4)$$

$$x_c^B = \frac{\dot{m}_c^B}{\sum_{c=1}^{N^C} \dot{m}_c^B} \quad \text{for } 1 \leq c \leq N^C, \quad (5)$$

$$x_c^D = \frac{\dot{m}_c^D}{\sum_{c=1}^{N^C} \dot{m}_c^D} \quad \text{for } 1 \leq c \leq N^C. \quad (6)$$

Using these flow rates and concentrations, we can calculate the reboiler and condenser duties using the following two equations:

$$Q^R = V^R \cdot (\hat{h}^V(\mathbf{x}^B) - \hat{h}^L(\mathbf{x}^B)), \quad (7)$$

$$Q^C = (D + L^C) \cdot (\hat{h}^L(\mathbf{x}^D) - \hat{h}^V(\mathbf{x}^D)). \quad (8)$$

Note that the reboiler duty Q^R will be non-negative and the condenser duty will be non-positive as long as we assume that the vapor enthalpy of a mixture at its dew point is larger than its liquid enthalpy at its bubble point.

We split the further reduction into two parts: one part solving the MESH equations for the stripping section and another part solving the MESH equations for the rectification section.

For the stripping section, we proceed upwards. In this procedure, we assume that for $n < N^F$, we have already calculated the values for $\mathbf{x}^n$. In the next step, we aim to calculate $\mathbf{y}^n, T^n, V^n, L^{n+1}$, and $\mathbf{x}^{n+1}$. We can use the mass balance and enthalpy balance around the bottom of the column up to stage n , combined with the equilibrium condition at stage n , to formulate the following MESH equations to be solved next:

$$V^n \mathbf{y}^n + (B + \bar{V}^R(n)) \mathbf{x}^B = L^{n+1} \mathbf{x}^{n+1}, \quad (9)$$

$$V^n \hat{h}^V(\mathbf{y}^n) + B \hat{h}^L(\mathbf{x}^B) + (\bar{V}^R(n)) \hat{h}^V(\mathbf{x}^B) = L^{n+1} \hat{h}^L(\mathbf{x}^{n+1}) + Q^R, \quad (10)$$

$$\mathbf{y}^n = \mathbf{K}(\mathbf{x}^n, \mathbf{y}^n, T^n, p) \mathbf{x}^n, \quad (11)$$

$$\sum_{c=1}^{N^C} x_c^{n+1} = \sum_{c=1}^{N^C} y_c^n = 1. \quad (12)$$

The control volume considered is shown in Fig. 3. We have a total of $2N^C + 3$ equations and $2N^C + 3$ unknowns that we want to determine.

Solving these equations gives us values for $\mathbf{x}^{n+1}$, allowing us to proceed to the next stage. By solving these equations, we also ensure that the mass balance and enthalpy balance for stage n hold, assuming that they already hold for stages 1 to $n-1$. For the first stage ($n=1$), we already know the values for $\mathbf{x}^1$, which are identical to $\mathbf{x}^B$.

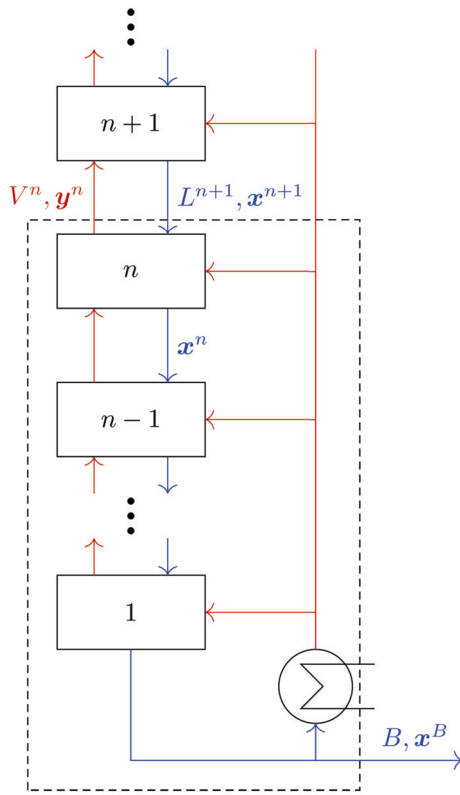


Fig. 3. Calculating the next variables upwards for stage n . The values of x^n are already known. The values L^{n+1} , x^{n+1} , T^n , y^n , V^n are calculated using the control volume depicted as a dashed box.

For the rectification section, we proceed downwards. Similarly, if we assume that for $n > N^F$, we already know y^n , we can calculate the values for T^n , x^n , L^n , V^{n-1} and y^{n-1} using the following equations:

$$L^n x^n + (D + \bar{L}^C(n)) x^D = V^{n-1} y^{n-1}, \quad (13)$$

$$L^n \hat{h}^L(x^n) + (D + \bar{L}^C(n)) \hat{h}^L(x^D) = V^{n-1} \hat{h}^V(y^{n-1}) + Q^C, \quad (14)$$

$$y^n = K(x^n, y^n, T^n, p) x^n, \quad (15)$$

$$\sum_{c=1}^{N^C} x_c^n = \sum_{c=1}^{N^C} y_c^{n-1} = 1. \quad (16)$$

Arriving at the feed stage N^F from both sides, we have calculated values for y^{N^F} and x^{N^F} from above and below. In general, they will not be in vapor-liquid equilibrium. Moreover, the mass balance and the enthalpy balance are not guaranteed to hold at the feed stage. To ensure that the MESH equations hold at every stage, including N^F , we need to solve the following equations in the outer loop:

$$y^{N^F} = K(x^{N^F}, y^{N^F}, T^{N^F}, p) x^{N^F}, \quad (17)$$

$$F x^F = B x^B + D x^D, \quad (18)$$

$$F h^L(x^F, T^F) + Q^R + Q^C = B \hat{h}^L(x^B) + D \hat{h}^L(x^D). \quad (19)$$

Note that we replaced the mass balance and enthalpy balance of stage N^F with the global balances.

The question is how to solve the introduced Equations (9)-(12) and (13)-(16). There are several important considerations. Firstly, the solution procedure needs to be fast since we will be solving these equations frequently. Additionally, it is crucial to achieve a high level of accuracy to avoid accumulating errors when calculating the next stage. Both can be achieved using the Newton method. However, a drawback of the Newton method is that its convergence is not guaranteed without safeguards. This is problematic because if we don't converge, we cannot

provide feedback to the outer loop, resulting in a failed reduction. To mitigate this issue, we will apply safeguarded solution methods, such as the bisection technique. As a byproduct of our considerations, we will also show that all concentrations and flow rates remain non-negative throughout the reduction process.

We start by considering the upwards calculations and consider a fixed stage $1 \leq n \leq N^F - 1$. We introduce an artificial variable defined as:

$$\alpha := \frac{V^n}{V^n + B + \bar{V}^R(n)}.$$

Our goal is to use this artificial variable to simplify the system of equations to just one equation. From the value for α , we can calculate the liquid composition x^{n+1} using the mass balance given in Equation (9) (divided by $V^n + B + \bar{V}^R(n)$):

$$x^{n+1} = \alpha y^n + (1 - \alpha) x^B.$$

Assuming that y^n and x^B are non-negative, and α is within the range $[0, 1]$, it follows that x^{n+1} will also be non-negative since it is a convex combination. Moreover, we can use α to calculate V^n and, once again using the mass balance, to determine L^{n+1} .

We can now divide the enthalpy balance (Equation (10)) by $V^n + B + \bar{V}^R(n)$ and obtain the following single equation to determine α :

$$\begin{aligned} \alpha \hat{h}^V(y^n) + \frac{B}{V^n + B + \bar{V}^R(n)} \hat{h}^L(x^B) + \frac{\bar{V}^R(n)}{V^n + B + \bar{V}^R(n)} \hat{h}^V(x^B) \\ = \left(\alpha + \frac{B + \bar{V}^R(n)}{V^n + B + \bar{V}^R(n)} \right) \hat{h}^L(x^{n+1}) + \frac{Q^R}{V^n + B + \bar{V}^R(n)}. \end{aligned} \quad (20)$$

Setting $\alpha = 0$ gives $V^n = 0$, $x^{n+1} = x^B$, $L^{n+1} = B$ and we can write:

$$\begin{aligned} \alpha \hat{h}^V(y^n) + \frac{B}{V^n + B + \bar{V}^R(n)} \hat{h}^L(x^B) + \frac{\bar{V}^R(n)}{V^n + B + \bar{V}^R(n)} \hat{h}^V(x^B) \\ = \frac{B}{B + \bar{V}^R(n)} \hat{h}^L(x^B) + \frac{\bar{V}^R(n)}{B + \bar{V}^R(n)} \hat{h}^V(x^B) \\ \leq \frac{B + \bar{V}^R(n)}{B + \bar{V}^R(n)} \hat{h}^L(x^B) + \frac{Q^R}{B + \bar{V}^R(n)} \\ = \left(\alpha + \frac{B + \bar{V}^R(n)}{V^n + B + \bar{V}^R(n)} \right) \hat{h}^L(x^{n+1}) + \frac{Q^R}{V^n + B + \bar{V}^R(n)} \end{aligned}$$

where the inequality follows from the calculation of Q^R given in Equation (7) and $\bar{V}^R(n) \leq V^R$. This means that the left-hand side of Equation (20) is less than or equal to the right-hand side. On the other hand, setting $\alpha = 1$ gives $V^n = \infty$, $x^{n+1} = y^n$, $L^{n+1} = \infty$ and we receive:

$$\begin{aligned} \alpha \hat{h}^V(y^n) + \frac{B}{V^n + B + \bar{V}^R(n)} \hat{h}^L(x^B) + \frac{\bar{V}^R(n)}{V^n + B + \bar{V}^R(n)} \hat{h}^V(x^B) \\ = \hat{h}^V(y^n) \\ \geq \hat{h}^L(y^n) \\ = \left(\alpha + \frac{B}{V^n + B + \bar{V}^R(n)} \right) \hat{h}^L(x^{n+1}) + \frac{Q^R}{V^n + \bar{V}^R(n)} \end{aligned}$$

where the inequality holds this time assuming that the vapor enthalpy of a composition at dew point is larger than its liquid enthalpy at bubble point. In summary, this means that for $\alpha = 0$, the left side is smaller than the right side and vice versa for $\alpha = 1$. Under the assumption of continuity of the model functions, this implies that there exists a solution $\alpha \in [0, 1]$, which can be found with bisection.

To determine a solution reliably and fast, we can use a safeguarded version of the secant method applied to the single equation. We can use

the values of two points to construct a secant. In the first iteration of the secant method, we can use the bounds of our initial search interval $[0, 1]$ and later on always use the last two iterates. Solving the linear equation based on the secant gives us a new iterate. Provided this iterate is contained within our current search interval, we can use this iterate to update the interval in which a solution must be contained. If the next iterate is outside the search interval, we can discard the iterate and use the midpoint of our search interval as the next iterate, thereby reducing the search interval. The bisection ensures that the iterates converge to a solution. Once the iterates are close enough to a solution, the super-linear convergence of the secant method will dominate, ensuring fast convergence.

After reevaluating the single equation approach, it becomes apparent that we can always ensure finding a solution where all calculated values are non-negative. For any $\alpha \in [0, 1]$, the flow rates V^n and L^{n+1} , as well as the composition $\mathbf{x}^{n+1}$, will be non-negative given that $\mathbf{y}^n$ is non-negative. By employing the safeguarded secant method, we can ensure convergence towards the solution that is guaranteed to exist. If we can also ensure that the solution of the equilibrium condition leads to non-negative $\mathbf{y}^{n+1}$, we can demonstrate inductively that all values remain non-negative. This guarantees the applicability of the reduction throughout the iterative outer loop. This is an important benefit over an approach using a simulation and varying the number of stages in an outer loop. It is in general hard to give guarantees for a simulation to converge. A failing simulation can then lead to a failing optimization.

For the downwards computations, we can derive a single equation using the following variable:

$$\beta := \frac{L^n}{L^n + D + \bar{L}^C(n)}.$$

By utilizing the mass balance, we can compute values for $\mathbf{y}^{n-1}$, V^{n-1} , and L^n . The enthalpy balance provides the following equation:

$$\begin{aligned} \beta \hat{h}^L(\mathbf{x}^n) + \frac{(D + \bar{L}^C(n))}{L^n + D + \bar{L}^C(n)} \hat{h}^L(\mathbf{x}^D) \\ = (\beta + \frac{(D + \bar{L}^C(n))}{L^n + D + \bar{L}^C(n)}) \hat{h}^V(\mathbf{y}^{n-1}) + \frac{Q^C}{L^n + D + \bar{L}^C(n)}. \end{aligned}$$

To show the existence of a solution, we can argue similarly to the upwards calculations. By letting $\beta = 0$ ($L^n = 0$, $\mathbf{y}^{n-1} = \mathbf{x}^D$) we can write:

$$\begin{aligned} \beta \hat{h}^L(\mathbf{x}^n) + \frac{D + \bar{L}^C(n)}{L^n + D + \bar{L}^C(n)} \hat{h}^L(\mathbf{x}^D) \\ = \hat{h}^L(\mathbf{x}^D) \\ \geq \hat{h}^V(\mathbf{x}^D) + \frac{Q^C}{D + \bar{L}^C(n)} \\ = (\beta + \frac{D + \bar{L}^C(n)}{L^n + D + \bar{L}^C(n)}) \hat{h}^V(\mathbf{y}^{n-1}) + \frac{Q^C}{L^n + D + \bar{L}^C(n)} \end{aligned}$$

where the inequality follows from the calculation of Q^C given in Equation (8) and $\bar{L}^C(n) \leq L^C$. On the other hand, by setting $\beta = 1$ ($L^n = \infty$, $\mathbf{y}^{n-1} = \mathbf{x}^n$) we can write:

$$\begin{aligned} \beta \hat{h}^L(\mathbf{x}^n) + \frac{(D + \bar{L}^C(n))}{L^n + D + \bar{L}^C(n)} \hat{h}^L(\mathbf{x}^D) \\ = \hat{h}^L(\mathbf{x}^n) \\ \leq \hat{h}^V(\mathbf{x}^n) \\ = (\beta + \frac{D + \bar{L}^C(n)}{L^n + D + \bar{L}^C(n)}) \hat{h}^V(\mathbf{y}^{n-1}) + \frac{Q^C}{L^n + D + \bar{L}^C(n)} \end{aligned}$$

This means that we can again guarantee the existence of a solution $\beta \in [0, 1]$.

The stage-to-stage calculations we described efficiently resolve the majority of the MESH equations, providing a function that calculates all process variables $\mathbf{u}$ from a few optimization variables $\mathbf{z}$. For the optimization procedures discussed in subsequent sections, it is beneficial not only to evaluate the map specified in Equation (2) but also to compute derivatives.

In the next step of the stage-to-stage calculations, we address the system of equations outlined in Equations (9) to (12), where we know already values for $\mathbf{u}(\mathbf{z}) := (\mathbf{x}^n, \bar{V}^R(n), B, \mathbf{x}^B, Q^R)$ and compute values for $\hat{\mathbf{u}}(\mathbf{z}) := (\mathbf{y}^n, T^n, V^n, L^{n+1}, \mathbf{x}^{n+1})$. We can summarize the equations as:

$$g(\mathbf{u}(\mathbf{z}), \hat{\mathbf{u}}(\mathbf{z})) = 0.$$

Derivatives for $\bar{V}^R(n)$, B , $\mathbf{x}^B$ and Q^R with respect to the optimization variables $\mathbf{z}$ can be easily computed using Equations (1), (3), (5) and (7). If we now assume that we also already computed the derivative of $\mathbf{x}^n$ with respect to $\mathbf{z}$, then we have the derivatives of the given variables $\frac{\partial \mathbf{u}(\mathbf{z})}{\partial \mathbf{z}}$. We can then use the implicit function theorem and the chain rule to obtain:

$$\frac{\partial \hat{\mathbf{u}}(\mathbf{z})}{\partial \mathbf{z}} = - \left(\frac{\partial g(\mathbf{u}(\mathbf{z}), \hat{\mathbf{u}}(\mathbf{z}))}{\partial \hat{\mathbf{u}}} \right)^{-1} \frac{\partial g(\mathbf{u}(\mathbf{z}), \hat{\mathbf{u}}(\mathbf{z}))}{\partial \mathbf{u}} \cdot \frac{\partial \mathbf{u}(\mathbf{z})}{\partial \mathbf{z}}.$$

This means that we especially obtain derivatives for $\mathbf{x}^{n+1}$ with respect to the optimization variables and we can continue with the next stage.

For the downwards computations, one can proceed analogously using Equations (13) to (16).

4. Column design problem

After introducing the reduction approach, we now show how it can be applied to a column design problem. We want to focus on the number of stages above and below the feed stage. This is why we assume that the principal separation has been determined. This means that the feed stream with its flow rate, composition, and temperature is known. The pressure in the column is given, and we know which components are to be withdrawn in the bottom and distillate streams. We also assume that the reflux ratio $\hat{r}^D$, i.e., the ratio between D and L^C , and the flow rates of the bottom and distillate streams are known. We denote the given bottom flow rate by $\hat{B}$ and the given distillate flow rate by $\hat{D}$. Because we cannot achieve a purity of 1 in the product streams, we instead demand that the purity in the streams is larger than a given value $\hat{x} \in [0, 1]$. We are interested in how many stages are needed to achieve this purity.

To include the specifications, we can add the following two equations to the optimization problem:

$$\begin{aligned} \sum_{c=1}^{N^C} \hat{m}_c^B &= \hat{B}, \\ L^C &= \hat{r}^D D. \end{aligned}$$

To make sure that the stage-to-stage computation described in Section 3 can be performed, we assume that $\hat{B} > 0$ and $\hat{D} > 0$.

As an additional constraint, we need to make sure that the demanded purity $\hat{x}$ is achieved. For a desired component c in, e.g., the distillate stream we could thus demand

$$x_c^D \geq \hat{x}.$$

However, depending on the reflux ratio, it might happen that the feasible set becomes empty by adding this constraint. To avoid infeasible optimization problems, we add an additional variable $t \geq 0$ and only demand

$$x_c^D + t \geq \hat{x}.$$

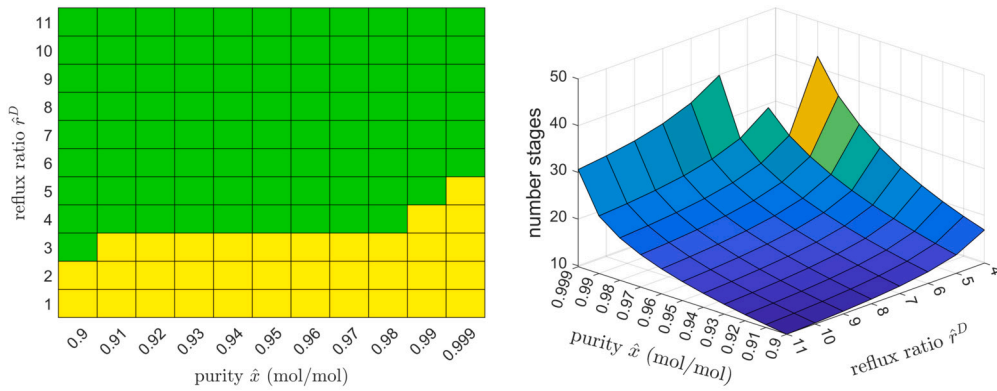


Fig. 4. Column design problem. Left picture: successful runs with achieved purity (green), successful runs without achieving purity (yellow), unsuccessful runs (red). Right picture: minimal number of stages for runs with achieved purities.

This way the constraint can always be satisfied. We add the additional variable with a large penalty parameter $\lambda > 0$ to the objective, to make sure that $t = 0$ whenever possible.

The introduced inequality makes sure that one component in the distillate is obtained with a high purity. Analogously, one can also demand a high purity in a bottom component. For the separation of multiple components, multiple inequalities can be added. Depending on the considered separation, other inequalities are possible. For example, one could also consider bounding the impurity in a product stream, i.e., the concentration of components that are to be separated from this stream.

As lower bounds, we need to assume that all flows are non-negative. For the stripping section, we need to assume that $\mu^B \in [1, N^F]$ and for the rectification section, we must have $\mu^D \in [N^F, N^S]$. We collect these bounds as lb and ub and write

$$lb \leq z \leq ub,$$

where we set formally $lb_i = -\infty$ and $ub_i = \infty$, if there is no lower or upper bound for the i -th optimization variable, respectively.

As an objective, we consider the effective height of the column. In summary, we obtain the following optimization problem:

$$\begin{aligned} \min_{z, t} \quad & \mu^D - \mu^B + 1 + \lambda t \\ \text{s.t.} \quad & h_i(\mathbf{w}(\mathbf{z})) = 0 \quad \text{for all } i \in I_{\text{red}} \\ & x_c^D + t \geq \hat{x} \\ & \sum_{c=1}^{N^C} \dot{m}_c^B = \hat{B} \\ & L^C = \hat{r}^D \hat{D} \\ & lb \leq z \leq ub \\ & 0 \leq t \end{aligned} \quad (21)$$

where $h_i, i \in I_{\text{red}}$ denotes the equations not solved by the reduction procedure (Equations (17) to (19)). The map $\mathbf{z} \mapsto \mathbf{w}$ denotes the reduction described in Section 3. The variables μ^D, μ^B, x_c^D are contained in $\mathbf{w}$.

As a concrete example, we consider the separation of a mixture of acetone (AC), chloroform (CF) and benzene (BE) at 1 bar. As feed, we consider a stream with 1 mol/s, a composition of $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ mol/mol, and we assume that the feed stream is at the bubble point for the pressure of 1 bar. The NRTL vapor-liquid equilibrium model and parameters of the mixture are given in Appendix B.

A separation into two pure streams is not possible for the given mixture, as a heavy boiling azeotrope between acetone and chloroform is formed at approximately $x_{AC} = 0.3454$ mol/mol acetone in the mixture. This means that as the light boiling component, acetone can be retrieved in the distillate. In the following, we thus fix the flow rate for the bot-

tom stream at 0.7 mol/s and demand a high purity (between 0.9 and 0.999 mol/mol) of acetone in the distillate.

The stage-to-stage calculations have been implemented in MATLAB[®], version 24.1.0 (R2024a) and precompiled using MATLAB[®] Coder[™]. The resulting optimization problems were solved using MATLAB[®] Optimization Toolbox[™]. To solve the problem given in Equation (21), we used the function `fmincon` with the standard settings, but also provided derivatives for the objective and the constraints computed analytically as described above in Section 3. This means that an interior point solver (see e.g. Nocedal and Wright (2006)) is used to solve the optimization problems.

To investigate the stability of our approach, we considered 11 different values for the desired purity

$$\hat{x} \in \{0.90, 0.91, 0.92, 0.93, 0.94, 0.95, 0.96, 0.97, 0.98, 0.99, 0.999\}$$

and varied the value of the reflux ratio $\hat{r}^D$ between 1 and 11 with 11 equidistant points. We set the maximal number of stages to $N^S = 51$ and the feed stage to $N^F = 26$. As initial points, we use typical values: $\mu^B = 12.5$, $\mu^D = 38.5$, $Bx^B = 0.7 \cdot (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, $Dx^D = 0.3 \cdot (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, $V^R = L^C = 5$. We set $\sigma = 1$ and the penalty parameters $\lambda = 10000$.

The results are shown in Fig. 4. In the left picture, we depict the different runs and show which of the runs have been solved successfully and which haven't (red). For the successful runs we distinguish between solutions that satisfy the purity constraint $x_{AC} \geq \hat{x}$ (green) and not (yellow). The plot shows that all calculations have been successful. However, not all solutions respect the purity constraint. Here $t > 0$ and there is no solution satisfying $x_{AC}^D \geq \hat{x}$. The runtime for each optimization varied between 0.67 and 11.87 seconds, and the mean was 2.63 seconds. The number of iterations varied between 15 and 47 with a mean of 26.67.

On the right, the effective number of stages is shown over the demanded purity $\hat{x}$ and the reflux ratio $\hat{r}^D$ for all points where the purity has been achieved. As expected, one can observe that the number of stages increases for larger purities $\hat{x}$ and a smaller reflux ratio. Starting for a reflux ratio of 5, one can see that the maximally allowed number of stages is not sufficient to achieve the demanded purity for high purities. Recalculating the point for $\hat{x} = 0.99$ and $\hat{r}^D = 4$ with $N^S = 81$ and $N^F = 41$ shows that 55 stages are needed.

This first numerical study shows that we find solutions to the column design problem reliably and fast. The question that arises is the quality of the calculated solutions, in the sense of how close they are to the global solution of the original mixed-integer non-linear optimization problem. To compare the solutions, as a first step, we round the continuous stage positions to a discrete stage position by taking the stage with the maximal flow to the stage, i.e.:

$$\arg \max \{V^R(n) \mid n = 1, \dots, N^F\}, \quad \arg \max \{L^C(n) \mid n = N^F, \dots, N^S\}.$$

Table 1

Rounded continuous solution (n^B, n^D) versus solution obtained using full enumeration ($n^{B,enum}, n^{D,enum}$).

$\hat{\rho}^D$	$\hat{x}$	μ^B	μ^D	x_{AC}^D	n^B	n^D	$x_{AC}^{D,r}$	$n^{B,enum}$	$n^{D,enum}$
3	0.95	1	51	0.9109	25	25	0.9063	25	25
4	0.95	10.3967	37.6538	0.95	16	12	0.9553	16/15/14	11/12/13
6	0.95	16.509	34.2103	0.95	9	8	0.9512	9/8	8/9
6	0.999	11.0966	51	0.999	15	25	0.999	15	25

By the construction using the Gaussian distribution, the stages will always be the means rounded to the nearest natural number, which we denote by $\lfloor \mu^B \rfloor, \lfloor \mu^D \rfloor$. We can then calculate the number of stages strictly below the stages in the feed stage, i.e., in the stripping section as:

$$n^B = N^F - \lfloor \mu^B \rfloor.$$

and the number of stages strictly above the feed stage, i.e., in the rectification section as:

$$n^D = \lfloor \mu^D \rfloor - N^F.$$

As we fixed in the column design problem the reflux ratio $\hat{\rho}^D$ and the bottom stream flow rate $\hat{B}$, we can now run a simulation for the obtained number of stages. As a first criterion, we can compare the obtained purity (in the following x_{AC}^r) with the purity of the continuous solution. For four exemplary points, the values are given in Table 1. As one can see in the table, the purity of the simulation with the rounded number of stages is for the considered feasible points slightly higher than the concentration of the continuous solution. This means that the rounded solutions are feasible for the discrete optimization problem.

As we can run a simulation for a fixed number of stages, we can, for a given $\hat{\rho}^D$ and demanded purity $\hat{x}$, also try out all possible combinations of stages above and below the feed stage (625 simulations). For all simulations, we can evaluate the objective given in Equation (21) and select the combination of stages with the smallest objective. We denote the number of stages in the stripping part by $n^{B,enum}$ and in the rectification part by $n^{D,enum}$. The results of the full enumeration are also given in Table 1 for the four points. It is possible that there are multiple solutions with the same total number of stages. In this case we list all solutions. For all points, we can observe that the difference between the solution obtained by rounding the continuous solution is very close to the solution obtained by the expensive full enumeration. For the second point, the solution differs by only one stage.

5. Flowsheet design

After demonstrating how this approach can be used for a single column example, we now turn to complete flowsheets with potentially several distillation columns. We will first describe how we can handle an arbitrary number of distillation columns. As a numerical example, we will consider the pressure swing distillation of acetone and chloroform.

In general, we can consider multiple distillation columns that we denote by $C_1, \dots, C_{N_K}$. To distinguish between the different columns, we add the column name as a superscript to the variables. This means that we consider the following optimization variables z :

- the component flow rates $\dot{m}_c^{B,C_k} := B^{C_k} x_c^{B,C_k}, \dot{m}_c^{D,C_k} := D^{C_k} x_c^{D,C_k}$ for $1 \leq c \leq N^C, 1 \leq k \leq N_K$.
- the mean positions μ^{B,C_k}, μ^{D,C_k} for $1 \leq k \leq N_K$.
- the flow rates V^{R,C_k}, L^{C,C_k} for $1 \leq k \leq N_K$.

We can perform the reduction for each column separately forming the global reduction map $z \mapsto w$. For each column $1 \leq k \leq N_K$ we still need to solve the equations

$$h_i^{C_k}(w(z)) = 0, \quad \text{for all } i \in I_{red}$$

described for a single column in Equations (17) to (19). Note that in the mass and enthalpy balances, product streams from different columns can serve as feed streams. If recycles are present in the flowsheets, they can be included as multiple feed streams for the column, resulting in multiple terms in the mass and enthalpy balances.

As before we can add the lower bound of 0 to ensure non-negativity. For μ^{B,C_k} and μ^{D,C_k} we can add lower and upper bounds to make sure that they are in the interval $[1, N^F]$ and $[N^F, N^S]$, respectively. We collect these bounds again as lb and ub and write

$$lb \leq z \leq ub.$$

To make sure that we can always apply the stage-to-stage calculation introduced in Section 3, we need to add bounds for the flow rate of each bottom and distillate stream. By introducing a minimal flow rate L_{min} , two inequalities can be added for each column $1 \leq k \leq N_K$

$$\sum_{c=1}^{N^C} \dot{m}_c^{B,C_k} \geq L_{min},$$

$$\sum_{c=1}^{N^C} \dot{m}_c^{D,C_k} \geq L_{min}.$$

For simplicity we denote the inequalities by $A \cdot z \leq b$.

The equations described so far make sure that each feasible point satisfies all MESH equations and is thus a valid simulation. Additionally, we can introduce an objective function f^{opt} that maps the process variables to a real value. Equality constraints h^{opt} and inequality constraints g^{opt} can be added as additional specifications.

Altogether, we can formulate the following optimization problem:

$$\begin{aligned} \min_z \quad & f^{opt}(w(z)) \\ \text{s.t.} \quad & h_i^{C_k}(w(z)) = 0 \quad \text{for all } i \in I_{red}, 1 \leq k \leq N_K \\ & h^{opt}(w(z)) = 0 \\ & g^{opt}(w(z)) \leq 0 \\ & A \cdot z \leq b \\ & lb \leq z \leq ub \end{aligned}$$

In a concrete example, we consider again the mixture of acetone and chloroform. To split the mixture into two pure components, we consider a pressure swing distillation with two distillation columns. The flowsheet is depicted schematically in Fig. 5. The feed (Stream 1) enters a first column which is operated at a pressure of 5 bar. The bottom stream of the first column enters the second column, and the top product is the product stream for acetone. The second column is operated at the pressure of 1 bar. The top stream is the product stream for chloroform. The bottom stream is recycled and mixed with the original feed. We assume that Stream 1 has a flow rate of 1 mol/s, the composition [0.5, 0.5] mol/mol, and is at bubble point temperature for 5 bar.

As objective, we aim at minimizing the sum of reboiler duties in both columns. We demand that the concentration of acetone in Stream 2 and the concentration of chloroform in Stream 4 are both above 0.95 mol/mol. We allow the optimizer to assign the stages freely, but we bound the total sum of the stages in both columns from above. As pa-

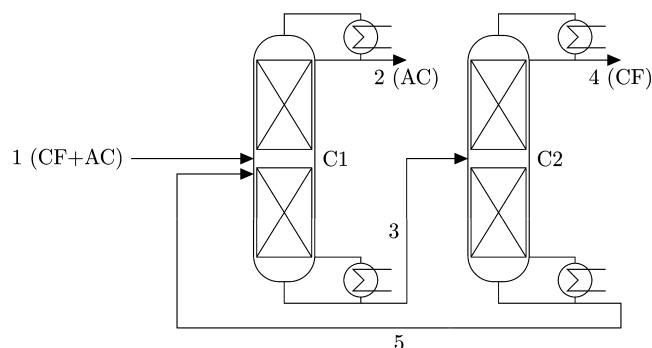


Fig. 5. Flowsheet of the pressure-swing distillation process for separating acetone (AC) and chloroform (CF).

Table 2

Variable stage distribution vs fixed stage distribution for pressure swing distillation.

	# stages	# iterations	# runtime	sum Q^R
variable stages	70	82	8.7684	355.85
	42	57	4.5003	628.76
given stages	70	64	6.6196	392.59
	42	60	4.1905	746.50

	# stages	μ^{B,C_1}	μ^{D,C_1}	μ^{B,C_2}	μ^{D,C_2}
variable stages	70	4.4017	49.9785	21.6114	44.0345
	42	9.3973	35.8798	16.2535	29.7710
given stages	70	18	52	18	52
	42	11	31	11	31

rameters, we set again $\sigma = 1$ and for the minimal allowed flow rate we choose $L_{\min} = 0.01$ mol/s.

As a comparison, we implemented an optimization with a fixed stage distribution, in which we distributed the number among both columns equally. Within the column, we assigned the same number of stages above and below the feed. Both optimizations have been solved using the same implementation as described in the previous section.

The results are shown in Table 2. We consider in the table two different cases: In the first one we bound the total number of stages by 70; in the second one by 42. The optimization with a free position of stages needs a similar number of iterations and is only slightly slower than the one for given stages. In the table, one can see that the optimized stage positions are not symmetric. In both cases the first column is larger than the second, e.g., 46.58 vs 23.42 stages for a bound of 70 stages. Also, within the column, the feed is not in the middle, but the number of stages in the stripping part is larger than the number of stages in the rectification part. This optimized design also leads to energy savings; while it is around 9.4% for the computation with at most 70 stages, it is already around 15.7% for 42 stages.

The results suggest that the energy savings with an optimized stage distribution are higher for a smaller number of stages. To confirm this observation on a larger range, we calculated two N-Q curves. One with an equal distribution of stages and one with an optimized distribution of stages. For the case of an equal distribution of stages, we start with 7 stages below and above the feed stage (total number of stages is 30). We minimize the reboiler duty for the fixed number of stages. We then increase the number of stages in each section by 1 and solve the optimization problem again. We continue until we reach 50 stages per section (total number of stages 202). The results are shown in Fig. 6. For the case of an optimized distribution of stages, we vary the total number of stages between 30 and 202 and minimize the reboiler duty for 100 equidistant points. The results are also shown in Fig. 6.

For a total number of stages of 30, the difference between both curves is relatively large. While with an optimized stage distribution, 1507 kW

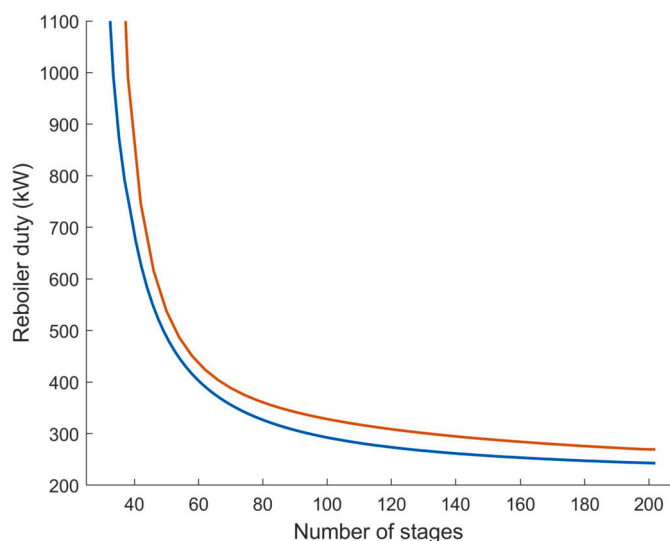


Fig. 6. N-Q curve for optimized stage distribution (blue) and uniform stage distribution (red).

are needed, the reboiler duty needed for a uniform distribution is 3711 kW (outside of the depicted range). For larger stage numbers, the difference becomes smaller. For a total number of stages of 202, the difference is only 243 kW versus 269 kW.

6. Conclusion

In the present work, we presented a continuous formulation for variable-sized distillation columns. The formulation considers the streams returning from the reboiler and the condenser to the columns as distributed streams. This way, we are able to avoid discrete degrees of freedom and can formulate a continuous nonlinear optimization problem. We use the full MESH equations in each stage. The model is thus highly nonlinear and can easily have thousands of variables per column. To reduce the model complexity, we introduced stage-to-stage calculations adapted to the distributed streams. We were able to show that the introduced reduction is well posed.

In Sections 4 and 5, we then applied the introduced concept first to the optimal design of a single column and then to a complete flowsheet. The numerical experiments showed the performance and stability of our approach. Without any initialization strategy, we were able to solve the column design problem over a wide range of specifications. The obtained results only differ by at most one stage compared to a solution obtained by full enumeration. The results for the flowsheet show the benefit of optimizing the stage distribution. Considerable energy savings can be realized, especially for a smaller number of stages.

In the present work, we assume a single feed position for a distillation column. In future work, a possible extension to multiple feed locations could be considered. Another interesting point is the maximally allowed number of stages in each section. If the number is too small, optimal solutions could be cut off. If there is a large number of stages, many almost empty stages may be considered, thus increasing the runtime. Potentially, an adaptive strategy could be combined with the strategies presented here.

CRedit authorship contribution statement

Tobias Seidel: Writing – original draft, Investigation, Conceptualization. **Lorenz T. Biegler:** Writing – review & editing, Supervision, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used FhGenie in order to correct the English language and improve readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. MESH equations for a distillation column

As mentioned in Section 2 we use the MESH equations for each stage.

• Mass balances:

for $n = 1, \dots, N_F - 1$ and $c = 1, \dots, N_C$:

$$V^n y_c^n + L^n x_c^n = V^{n-1} y_c^{n-1} + L^{n+1} x_c^{n+1} + V^R(n) x_c^B,$$

for $n = N_F + 1, \dots, N_S$ and $c = 1, \dots, N_C$:

$$V^n y_c^n + L^n x_c^n = V^{n-1} y_c^{n-1} + L^{n+1} x_c^{n+1} + L^C(n) x_c^D,$$

for $n = N_F$ and $c = 1, \dots, N_C$:

$$V^n y_c^n + L^n x_c^n = V^{n-1} y_c^{n-1} + L^{n+1} x_c^{n+1} + L^C(n) x_c^D + V^R(n) x_c^B + F x_c^F.$$

Reboiler and condenser:

$$L^1 = V^R + B,$$

$$x_c^1 = x_c^B, \quad \text{for } c = 1, \dots, N_C,$$

$$V^{N^S} = L^C + D,$$

$$y_c^{N^S} = x_c^D, \quad \text{for } c = 1, \dots, N_C.$$

For the first stage $n = 1$ there is no vapor stream from a stage below entering. The corresponding term in the equation can be dropped. The same holds for the liquid stream entering from above to the last stage $n = N^S$.

• Equilibrium conditions:

$$y_c^n = K_c(x^n, y^n, T^n, p) x_c^n \quad \text{for } i = 1, \dots, N_C, n = 1, \dots, N_S.$$

For the considered example, the vapor phase is treated as ideal. For the non-ideality of the liquid phase the NRTL model is used Renon and Prausnitz (1968), which means that the equilibrium ratios in the example are for $1 \leq c \leq N^C$ given as:

$$K_c(x, y, T, p) = \frac{p_c^V(T^n) \gamma_c(x, T)}{p},$$

where $p_c^V(T)$ is the vapor pressure of the pure component c at temperature T and $\gamma_c(x, T)$ is the activity coefficient of component c for the present composition x of the liquid phase at temperature T .

• Summation equations

$$\sum_{c=1}^{N_C} x_c^n = 1, \quad \text{and} \quad \sum_{c=1}^{N_C} y_c^n = 1, \quad \text{for } n = 1, \dots, N_S.$$

• Energy balances

for $n = 1, \dots, N_F - 1$:

$$L^{n+1} \hat{h}^L(x^{n+1}) + V^{n-1} \hat{h}^V(y^{n-1}) + V^R(n) \hat{h}^V(x^B)$$

$$= L^n \hat{h}^L(x^n) + V^n \hat{h}^V(y^n),$$

for $n = N_F$:

$$F \hat{h}^L(x^F, T^F) + L^{n+1} \hat{h}^L(x^{n+1}) + V^{n-1} \hat{h}^V(y^{n-1})$$

$$+ V^R(n) \hat{h}^V(x^B) + L^C(n) \hat{h}^L(x^D)$$

$$= L^n \hat{h}^L(x^n) + V^n \hat{h}^V(y^n),$$

for $n = N_F + 1, \dots, N_S$:

$$L^{n+1} \hat{h}^L(x^{n+1}) + V^{n-1} \hat{h}^V(y^{n-1}) + L^C(n) \hat{h}^L(x^D)$$

$$= L^n \hat{h}^L(x^n) + V^n \hat{h}^V(y^n),$$

Reboiler and condenser :

$$\dot{Q}_R + V^R \hat{h}^L(x^B) = V^R \hat{h}^V(x^B),$$

$$\dot{Q}_C + (D + L^C) \hat{h}^V(x^D) = (D + L^C) \hat{h}^L(x^D).$$

As for the mass-balance the enthalpy from the vapor stream below the first stage $n = 1$ and the enthalpy of the liquid stream above the last stage $n = N^S$ can be dropped.

Note that we have to explicitly model the reboiler and the condenser here. Often they are simply merged with a first and the last stage. However, we need the vapor stream produced by the reboiler and the liquid stream produced by the condenser and distribute them over the column.

Appendix B. Physical property models

The pure component properties required for the simulation are: vapor pressure, liquid enthalpy and vapor enthalpy. The models for these properties are taken from the DIPPR Database (DIPPR Project 801 (2005)) for the substances acetone, and chloroform.

We model the vapor enthalpy as convex combination of the single component vapor enthalpies h_c^V :

$$h^V(x, T) = \sum_c^{N^C} x_c \cdot h_c^V(T).$$

In the case that a mixture is at dew point, we can compute the temperature T and write:

$$\hat{h}^V(x) := h^V(x, T).$$

The vapor enthalpy is modeled as:

$$h_c^V(T) = A \cdot T + B \cdot \frac{C}{\tanh\left(\frac{C}{T}\right)} - D \cdot E \cdot \tanh\left(\frac{E}{T}\right).$$

The parameters are given in Table B.3.

The liquid enthalpy is modeled indirectly as the difference of the vapor enthalpy and the vaporization enthalpy:

$$h^L(x, T) = \sum_c^{N^C} x_c \cdot (h_c^V(T) - h_c^{VL}(T)).$$

In the case that a mixture is at bubble point, we can compute the temperature T and write:

$$\hat{h}^L(x) := h^L(x, T).$$

The vaporization enthalpy is modeled as:

$$h_c^{VL}(T) = A \cdot \left(1 - \frac{T}{F}\right)^{B + \frac{T}{F} \cdot \left(C + \frac{T}{F} \cdot \left(D + E \cdot \frac{T}{F}\right)\right)}.$$

The parameters are given in Table B.4.

The vapor pressure for a component c is modeled as

$$p_c^V(T) = \exp\left(A + \frac{B}{T} + C \cdot \log(T) + D \cdot T^E\right).$$

The parameters are given in Table B.5.

Table B.3

Parameters for vapor enthalpy.

Component	A	B	C	D	E
Acetone	56957.69	163081.7	764.7302	-77583.94	853.5851
Chloroform	39322.47	65749.01	909.0312	47893.78	390.2708
Benzene	43535.144	230323.05	683.07662	-82982.783	788.0827

Table B.4

Parameters for the vaporization enthalpy.

Component	A	B	C	D	E	F
Acetone	50272875	1.1628608	-1.4935691	0.74575418	0	508.2
Chloroform	41860000	0.3584	0	0	0	536.4
Benzene	50414286	0.66248342	-0.25822401	0	0	562.16

Table B.5

Parameters for the vapor pressure.

Component	A	B	C	D	E
Acetone	57.493074535	-5599.6	-7.0985	6.2237e-06	2
Chloroform	57.493074535	-5599.6	-7.0985	6.2237e-06	2
Benzene	71.5940745355	-6486.2	-9.2194	6.9844e-06	2

Table B.6

NRTL model parameters for the description of the vapor-liquid equilibrium.

Component i Component j	Acetone Chloroform	Acetone Benzene	Benzene Chloroform
a_{ij}	0.9646	-0.1015	0
a_{ji}	0.5382	0.4224	0
b_{ij}	-590.026	306.0663	-375.4311
b_{ji}	-106.4216	-239.9009	313.0115
α_{ij}	0.3	0.3	0.47

As mentioned before in Appendix A the vapor phase is treated as ideal. For the non-ideality of the liquid phase the NRTL model is used Renon and Prausnitz (1968):

$$\ln \gamma_1 = x_2^2 \left[\tau_{21} \left(\frac{G_{21}}{x_1 + x_2 G_{21}} \right)^2 + \frac{\tau_{12} G_{12}}{(x_2 + x_1 G_{12})^2} \right]$$

$$\ln \gamma_2 = x_1^2 \left[\tau_{12} \left(\frac{G_{12}}{x_2 + x_1 G_{12}} \right)^2 + \frac{\tau_{21} G_{21}}{(x_1 + x_2 G_{21})^2} \right]$$

with $G_{ij} = \exp(-\alpha \tau_{ij})$. We use the temperature relation $\tau_{ij} = a_{ij} + \frac{b_{ij}}{T}$. The parameters are given in Table B.6

Data availability

Data will be made available on request.

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